Attribute	Details
University / Institute	L J University — [IMCA]
Semester / Division	Sem-3 | Div-D
Subject	UDP
Group No.	Group-14
Back End Designer	Ayush Vyas (D-54)
Front End Designer	Rohit Pareek (D-58)
Database Designer	Het Moradiya (D-24)

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DOCUMENTATION
PROJECT : HOTEL ELITE
DEFINATION: HOTEL MANAGEMENT SYSTEM

1. INTRODUCTION

1.1 Existing System

Manual Registers: All guest details, room bookings, and payments are written by hand in paper notebooks or diaries.

Risk of Errors: High chance of human mistakes, such as booking the same room for two different guests at the same time

Time-Consuming: Check-in and check-out processes take a lot of time because staff must fill out everything manually on paper.

Communication Gap: The front desk has to call or physically walk to the housekeeping staff to check if a room is clean or dirty.

1.2 Need for the New System

Centralized Data: To manage all hotel activities like bookings, bills, staff logs, and kitchen orders from a single computer screen.

Fast & Paperless: To save time and storage space by storing records digitally instead of using paper registers.

Real-Time Tracking: To see the live status of every room instantly on a screen.

Online Booking: To allow customers to view room photos, check pricing, and book their rooms from anywhere in the world.

1.3 Objective of the New System

Automation: To automate room pricing based on seasons or weekends and calculate the final bill instantly.

Role-Based Security: To protect data by creating password-secured pages for Admin, Receptionist, and Housekeeping. No user can view pages outside their role.

Business Analytics: To show visual sales and profit graphs to the hotel owner to help them understand daily, weekly, and monthly growth.

1.4 Problem Definition

"The traditional manual system used in hotels causes booking mistakes, money calculation errors, and poor communication among staff. This makes customers unhappy and reduces hotel revenue. To solve these problems permanently, we are creating 'Hotel Elite', a fast, secure, and fully digital web application."

1.5 Core Components

Room Management: Manages room types and tracks their live status .

Booking & Reservation: Displays an easy calendar view to manage both online website bookings and walk-in customers.

Guest Management: Stores guest information safely, including names, phone numbers, ID cards, and past visit history.

Billing & Invoicing: Combines room prices, food bills, and taxes into one final, professional receipt.

Food & Room Service: Customers scan a QR code inside the room with their phone, browse the menu, and order food straight to the kitchen.

Housekeeping: Shows a clean-up checklist for the staff. Once they finish cleaning, they change the status back to 'Available'.

Reports & Analytics: Creates simple, colorful graphs for the Admin to track hotel income and customer feedback.

1.6 Project Scope :

In-Scope:Room booking, customized dashboards for Admin/Receptionist/Housekeeping, QR code scanning from rooms to order food, and automatic bill generation.

Out-of-Scope : It does not connect to real bank payment systems . It is designed for only one hotel, not for multiple hotel chains.

1.7 Assumptions and Constraints

Assumptions: The hotel will always have a stable internet connection so that staff devices can stay connected.

Constraints: The system needs internet to function. It is designed to fit within the short timeline of the college semester.

1.8 Advantages and Limitations

Advantages: Zero double-booking mistakes, lower paper costs, faster operational speed, and 100% accurate financial calculations.

Limitations: Staff members require basic training to use a computer or smartphone. The application stops working during total network outages.

2.PROJECT PLANNING

2.1 Team Details and Roles

Back End Designer - [D54 Ayush Vyas]:

Role: Responsible for backend development, server logic, and connecting the application with the database.

Front End Designer - [D58 Rohit Pareek]:

Role: Responsible for designing user interfaces, website layouts, buttons, and screen designs.

Database Designer - [D24 Het Moradiya]:

Role: Responsible for creating database schemas, managing data tables, and defining data relationships.

3. REQUIREMENT DETERMINATION & ANALYSIS

3.1 Requirement Determination

Functional Requirements: Secure login system, room status updates, bill generation, QR code food ordering, and report charts.

Non-Functional Requirements: Fast speed , high security , and a fully responsive screen design for mobiles and computers.

3.2 Targeted Users

Admin : Has full access. Manages staff accounts, updates room rates, and checks business profit reports.

Receptionist: Has medium access. Checks room availability, handles customer check-in/check-out, and prints final bills.

Housekeeping Staff: Has limited access. Reviews the dirty room list and updates status to 'Available' once cleaning is finished.

Customer: Has public access. Browses room pictures, checks available dates, books a room, and orders food inside the hotel.

4. SYSTEM DESIGN

4.1 Use Case Diagram

Customer: Browse Rooms, Book Rooms, Order Food.

Receptionist: Check-In Guests, Print Invoice, Check Available Rooms.

Housekeeping: View Dirty Rooms, Mark Rooms as Clean.

Admin: View Profit Analytics, Create Staff Accounts, plus all features above.

4.2 Data Dictionary

Table 1: admin

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique admin identification number
username	VARCHAR(50)	Not Null	Admin username for login
password	VARCHAR(255)	Not Null	Admin account password

Table 2: users

Column Name	Data Type	Constraint	Description

id	INT(11)	Primary Key, Auto Increment	Unique user identification number
name	VARCHAR(100)	Not Null	Full name of the user
email	VARCHAR(255)	Unique, Not Null	User email address for login
password	VARCHAR(255)	Not Null	User account password

Table 3: rooms

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique room ID
room_no	VARCHAR(20)	Not Null	Room number (e.g., 101, 102)
floor	VARCHAR(50)	Nullable	Floor location
room_type	VARCHAR(50)	Nullable	Standard Room, Deluxe Room, etc.

room_status	VARCHAR(50)	Default: 'Clean & Ready'	Room cleaning/readiness status
price	DECIMAL(10,2)	Not Null	Price per night stay
bed_type	VARCHAR(100)	Nullable	Type of bed in the room
image	VARCHAR(255)	Nullable	Image URL for the room
max_adults	INT(11)	Default: 2	Maximum adult occupancy
max_children	INT(11)	Default: 1	Maximum children occupancy
extra_adult_charge	DECIMAL(10,2)	Default: 500.00	Extra charge per additional adult
extra_child_charge	DECIMAL(10,2)	Default: 300.00	Extra charge per additional child

Table 4: room_bookings

Column Name	Data Type	Constraint	Description

id	INT(11)	Primary Key, Auto Increment	Unique reservation number
phone	VARCHAR(15)	Nullable	Contact phone number
booking_date	TIMESTAMP	Default: CURRENT_TIMESTAMP	Booking creation time
room_id	INT(11)	Nullable	Reference ID of the booked room
check_in	DATETIME	Nullable	Scheduled check-in date and time
check_out	DATETIME	Nullable	Scheduled check-out date and time
user_email	VARCHAR(255)	Nullable	Guest email address
user_name	VARCHAR(255)	Nullable	Guest full name

total_price	DECIMAL(10,2)	Nullable	Total room rent price
payment_method	VARCHAR(50)	Default: 'Cash'	Mode of payment
payment_status	VARCHAR(50)	Default: 'Pending'	Bill payment status

Table 5: food_items

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique food item ID
food_name	VARCHAR(255)	Not Null	Name of the food item
price	DECIMAL(10,2)	Not Null	Price of the food item

Table 6: food_orders (Room Service)

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique food order

			tracking number
customer_name	VARCHAR(100)	Nullable	Guest name
room_no	VARCHAR(20)	Nullable	Room number delivering to
items	TEXT	Nullable	Names and details of ordered food items
total_amount	INT(11)	Nullable	Total price of the food order
payment_mode	VARCHAR(50)	Nullable	Mode of payment (e.g., Online, Cash)
order_date	TIMESTAMP	Default: CURRENT_TIMESTAMP	Order placement time

status	VARCHAR(50)	Default: 'Ordered'	Kitchen preparation status
payment_status	VARCHAR(50)	Default: 'Pending'	Food bill payment status

Table 7: contact_messages

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique message ID
name	VARCHAR(100)	Not Null	Name of the person contacting
email	VARCHAR(100)	Not Null	Email address of the sender
message	TEXT	Not Null	Message body content
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Message submission timestamp

Table 8: staff

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique staff member ID
name	VARCHAR(100)	Not Null	Staff member full name
email	VARCHAR(100)	Unique, Not Null	Staff login email
password	VARCHAR(255)	Not Null	Staff account password
role	VARCHAR(50)	Default: 'Staff'	Job role (Manager, Receptionist, etc.)
work_assigned	TEXT	Nullable	Assigned duties description
status	VARCHAR(50)	Default: 'Active'	Current working status
Created At	TimeStamp		

5. PROPOSED ENHANCEMENTS

AI Chatbot: An automated helper tool on the website to answer customer questions about pricing and room features immediately.

IoT Smart Room Keys: A digital key code sent directly to the customer's phone to open room doors without a physical metal key.

6. CONCLUSION

The Hotel Elite web application replaces slow, error-prone manual hotel records with a single, synchronized platform. It blocks double-booking errors, saves staff time, protects customer information, and provides a clear financial breakdown to the hotel owner.

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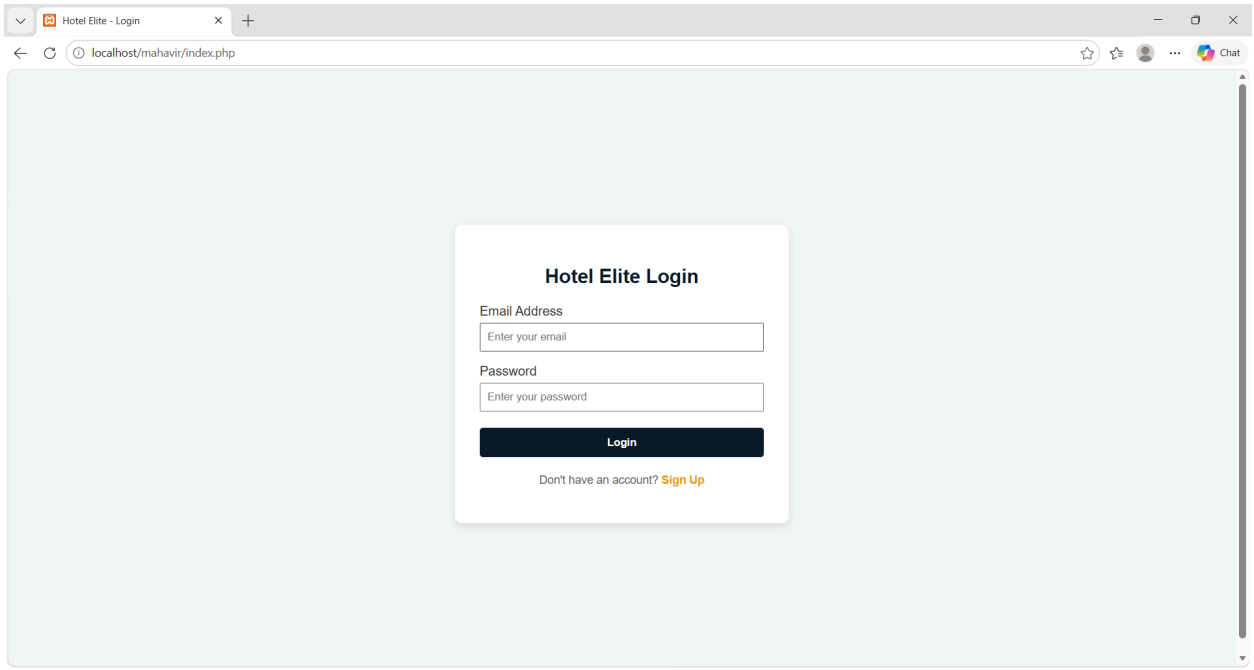
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Institution:L J university

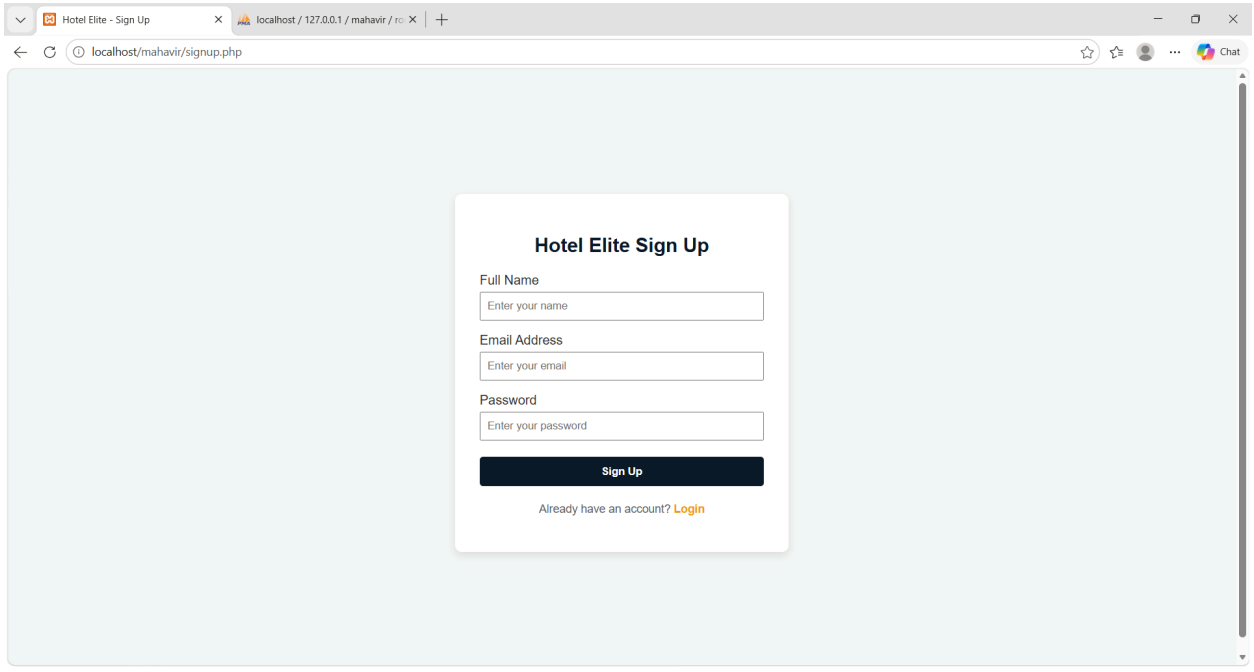
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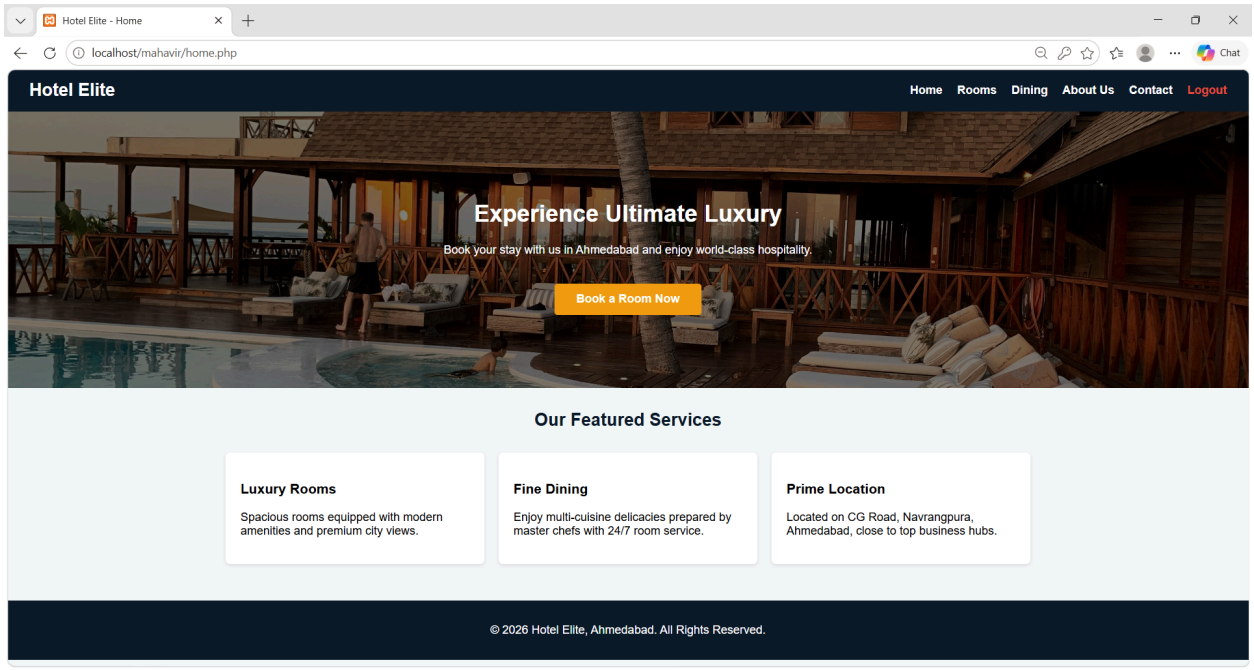
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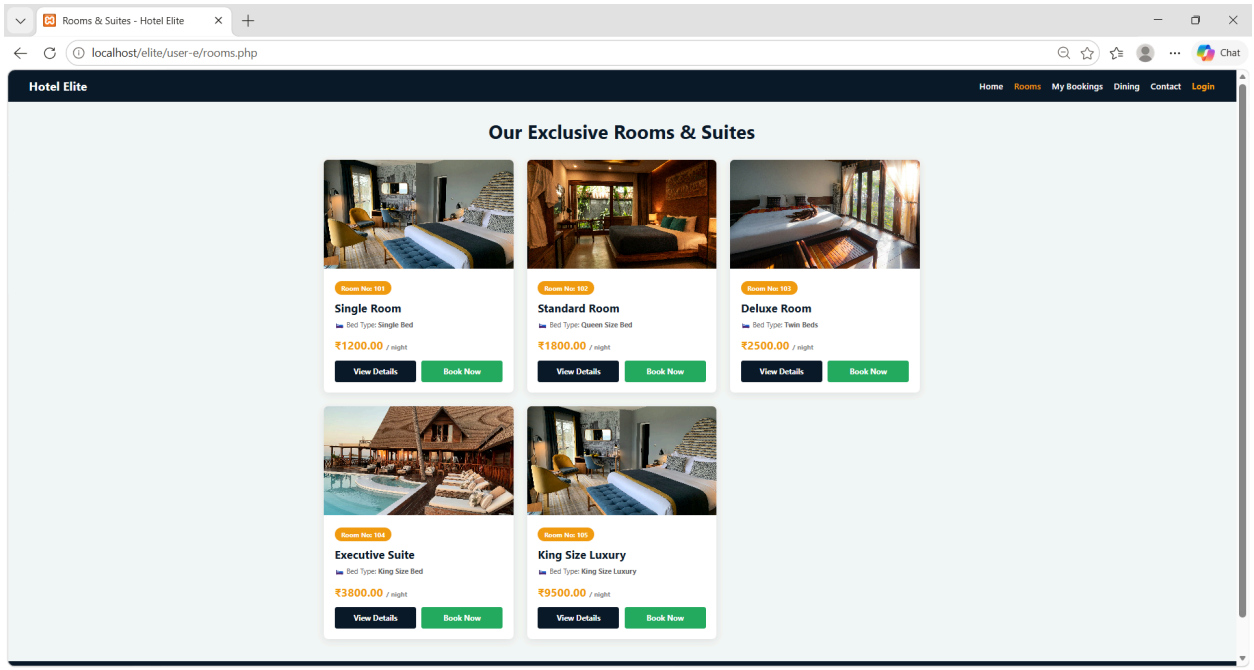
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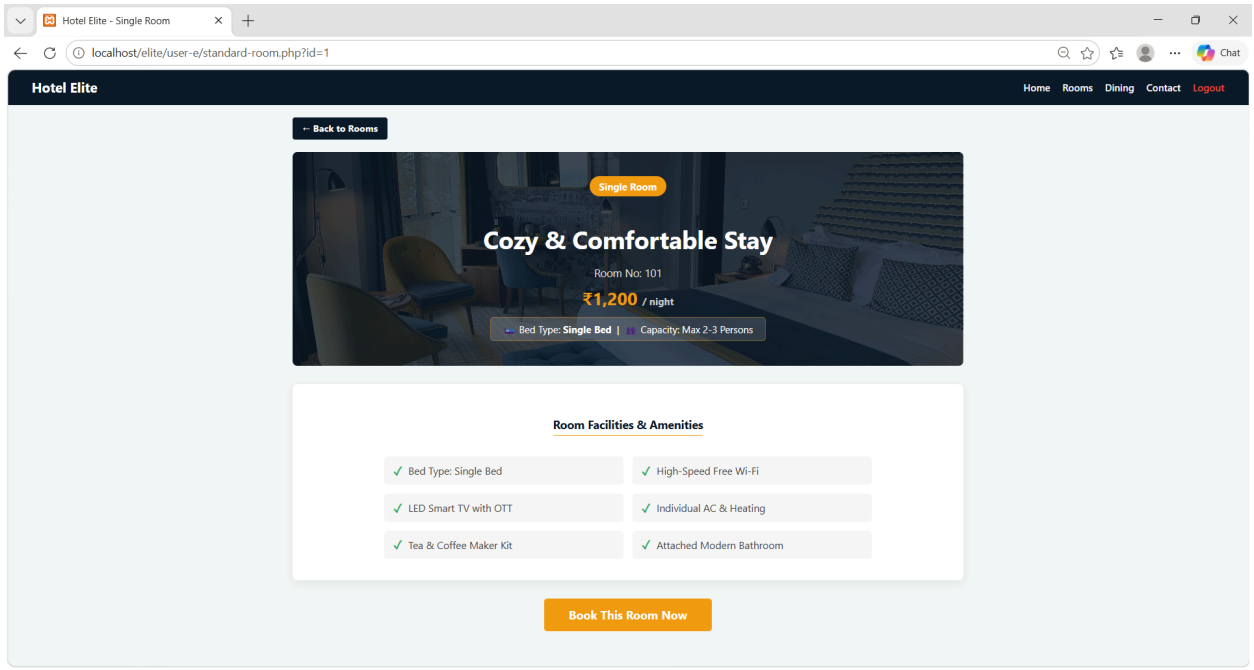
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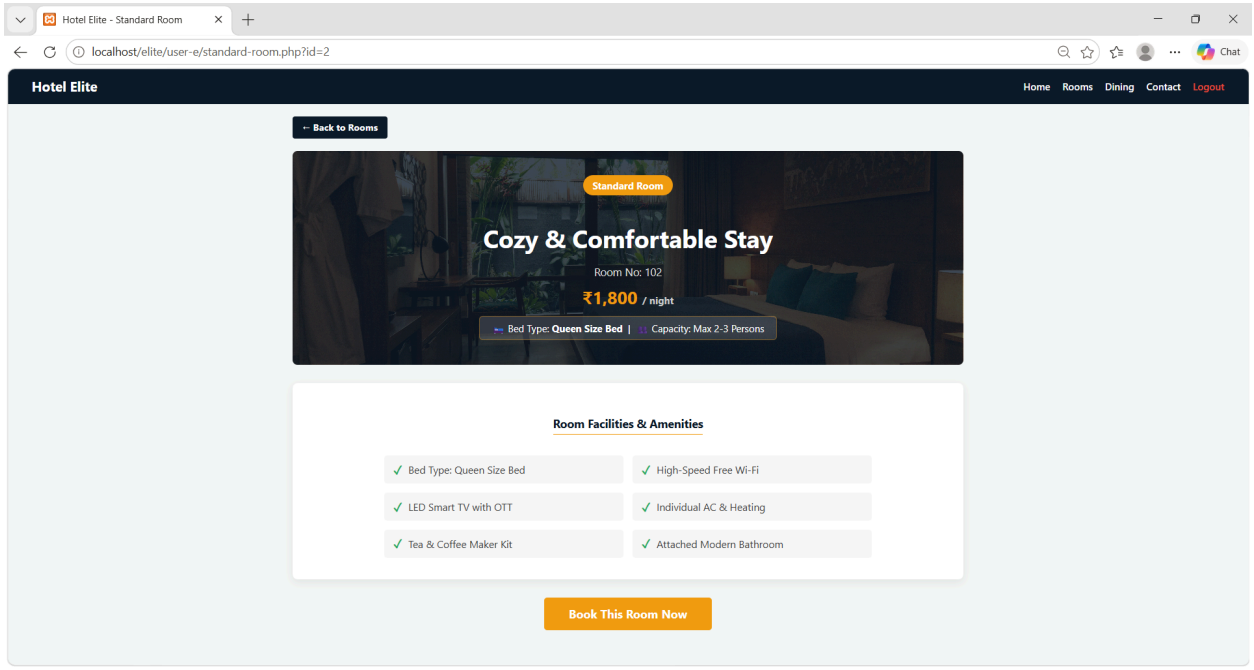
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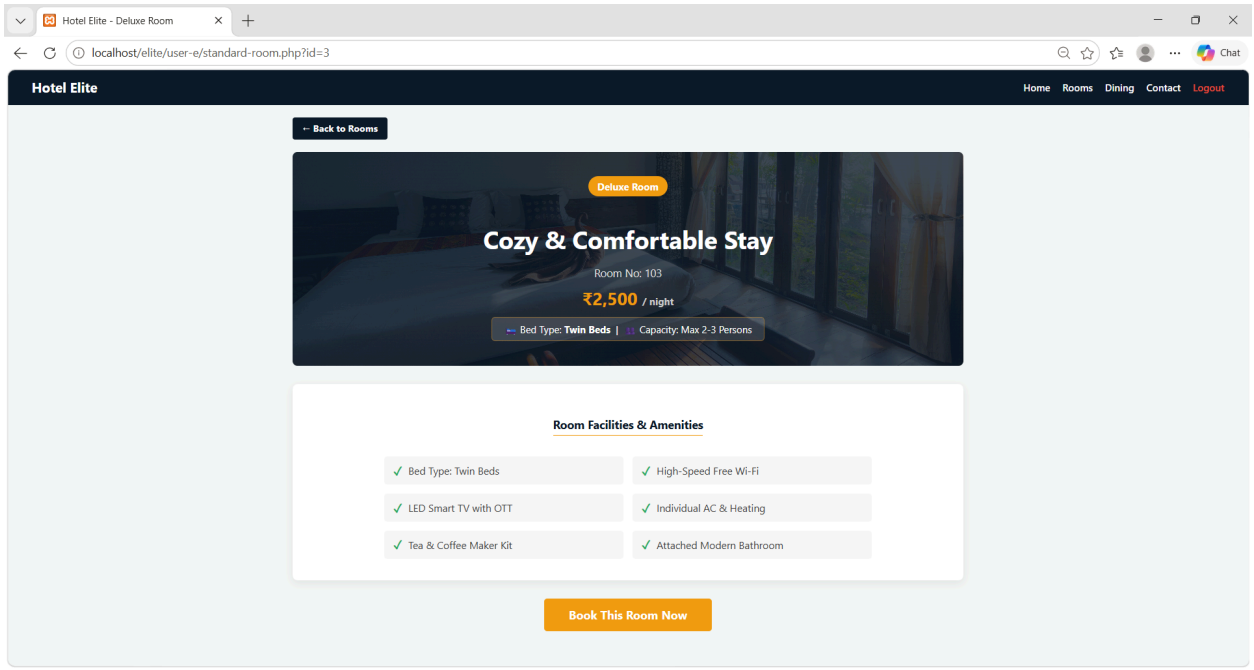
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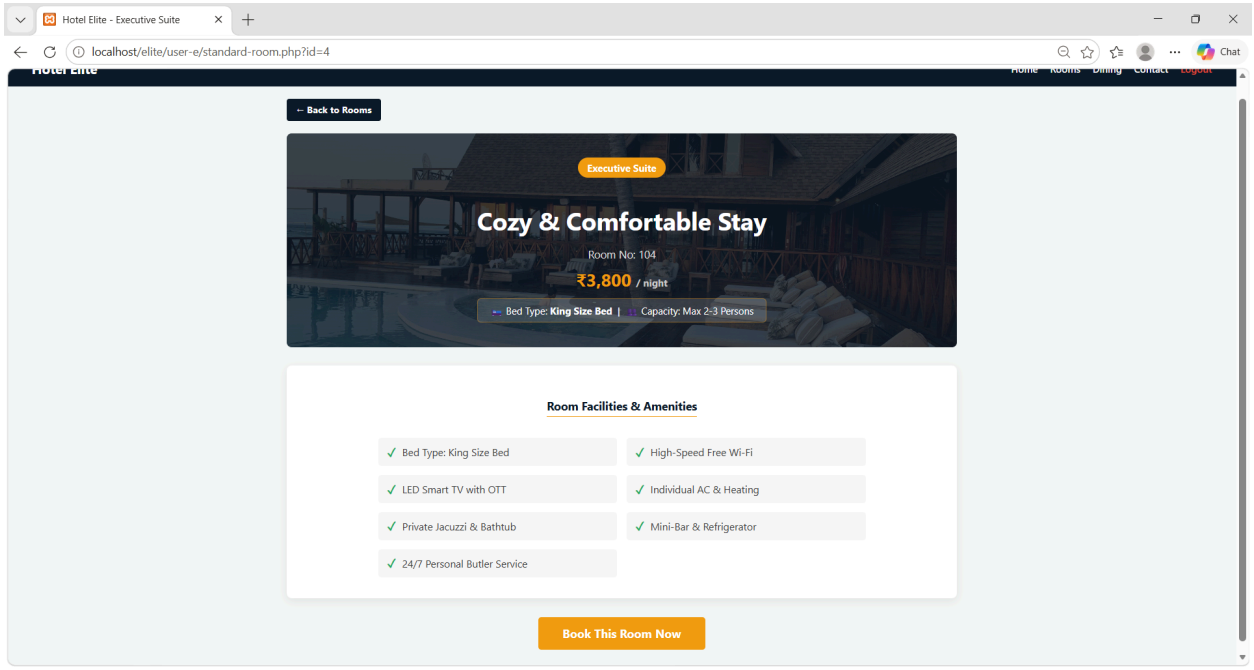
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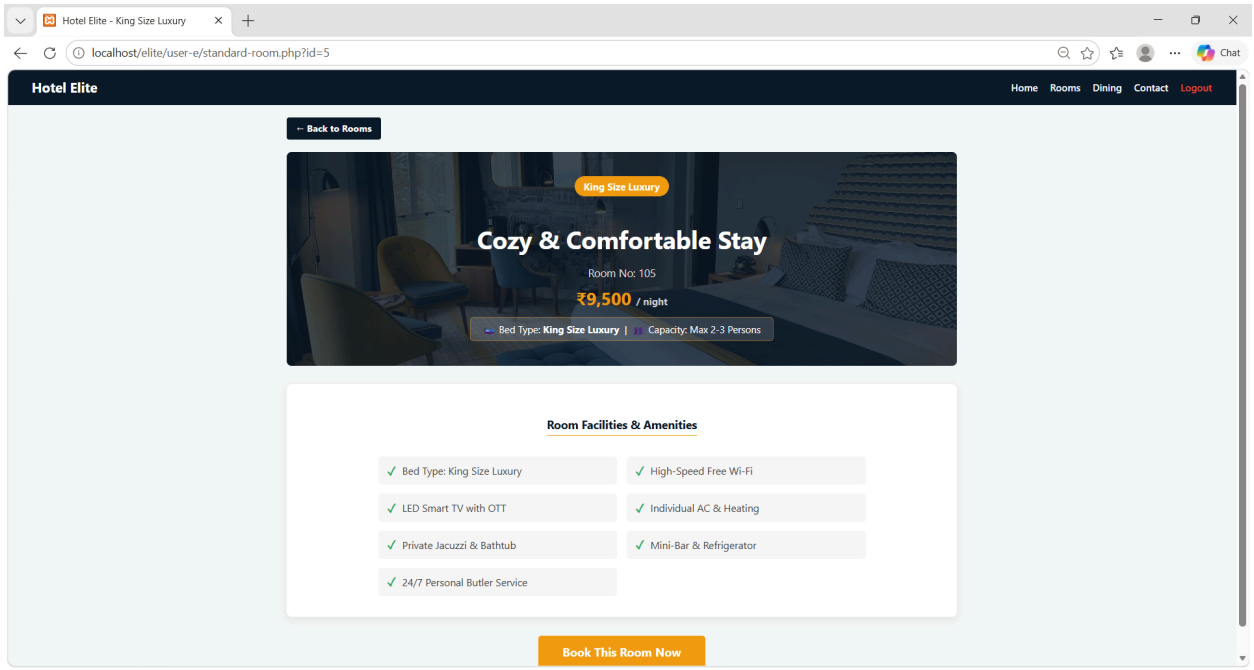
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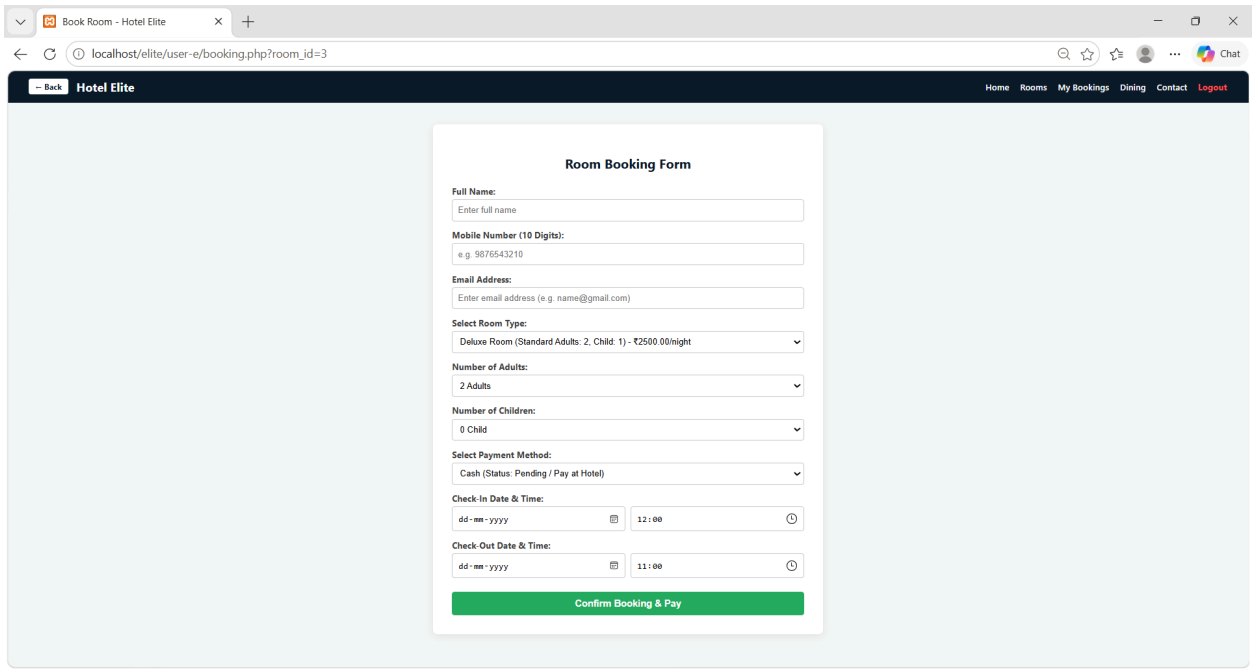
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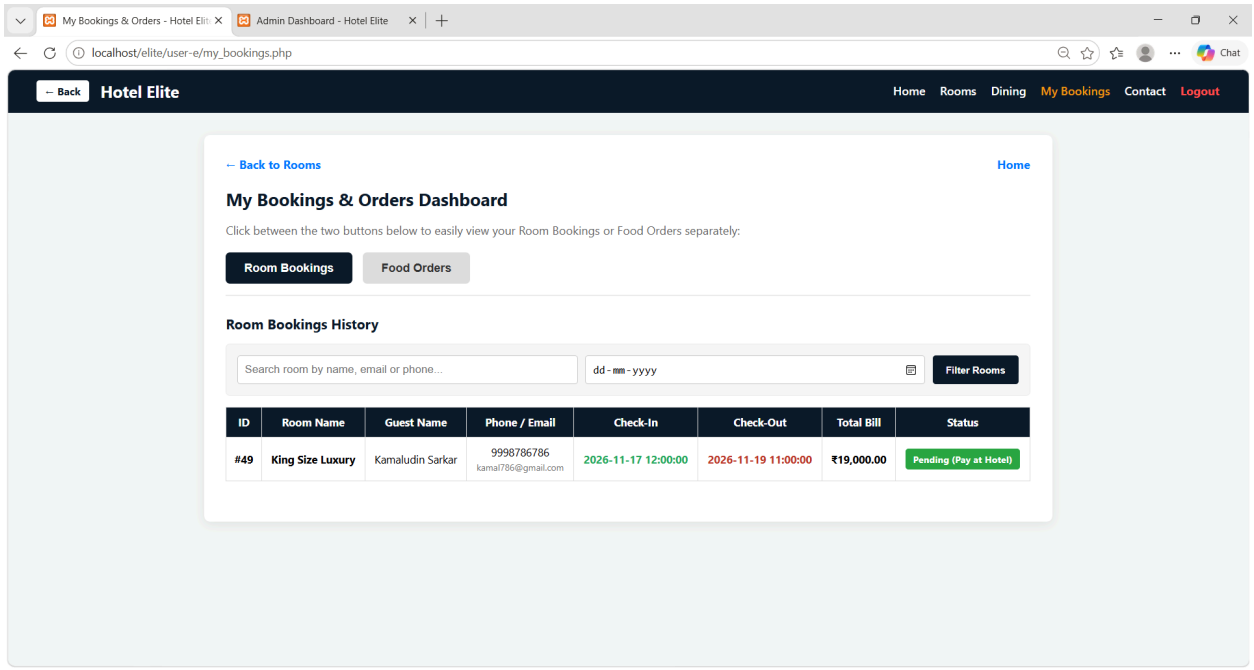
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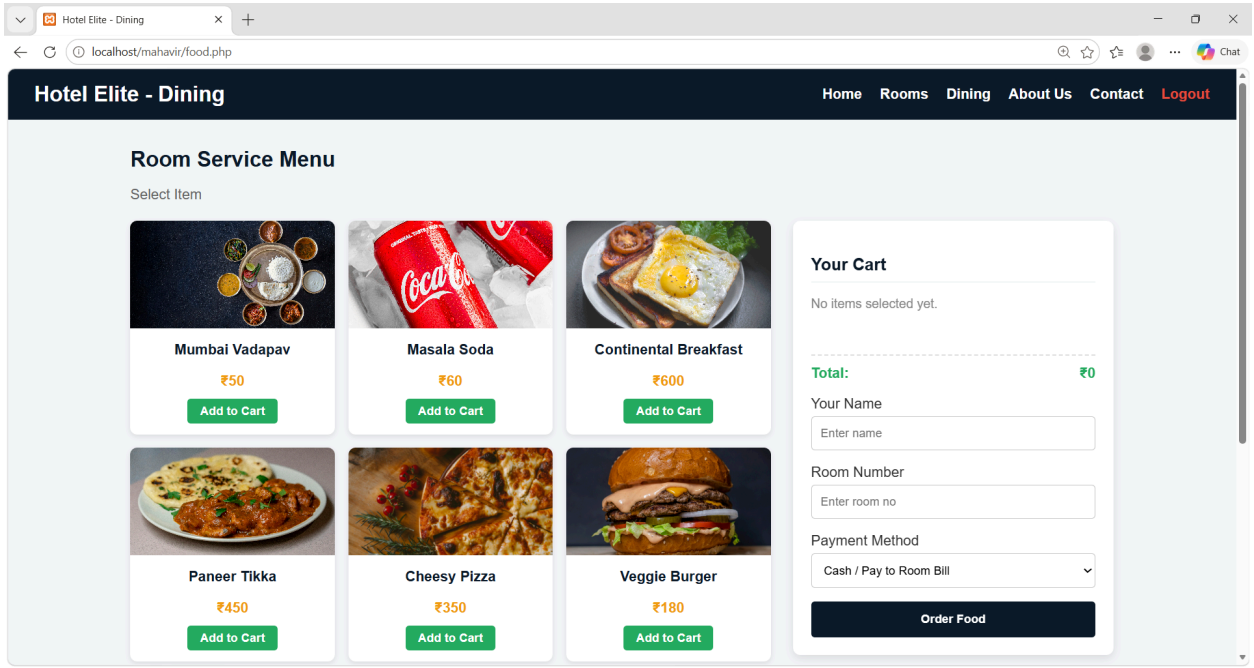
BOOK THIS ROOM

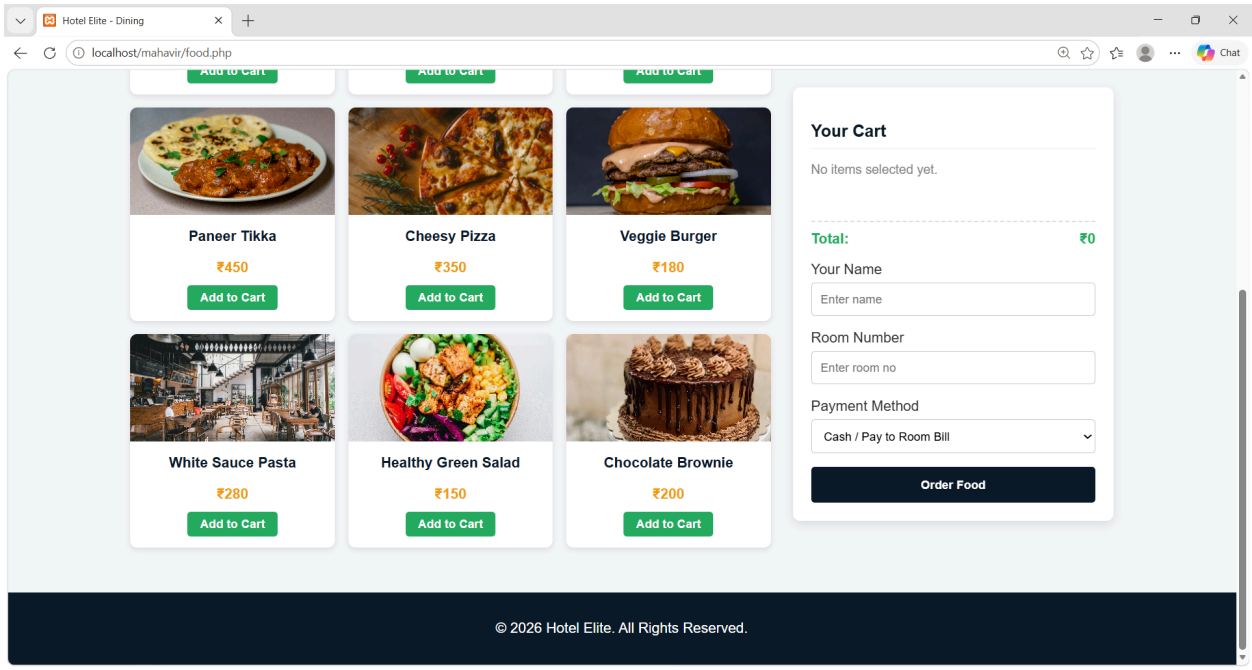


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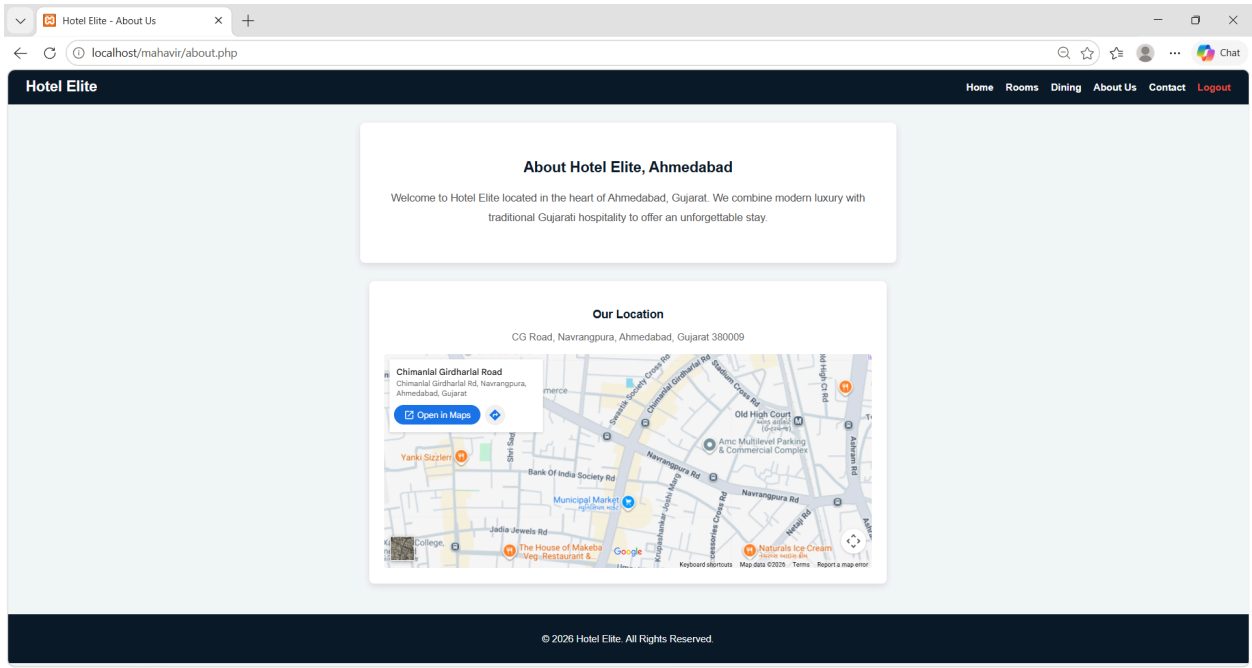


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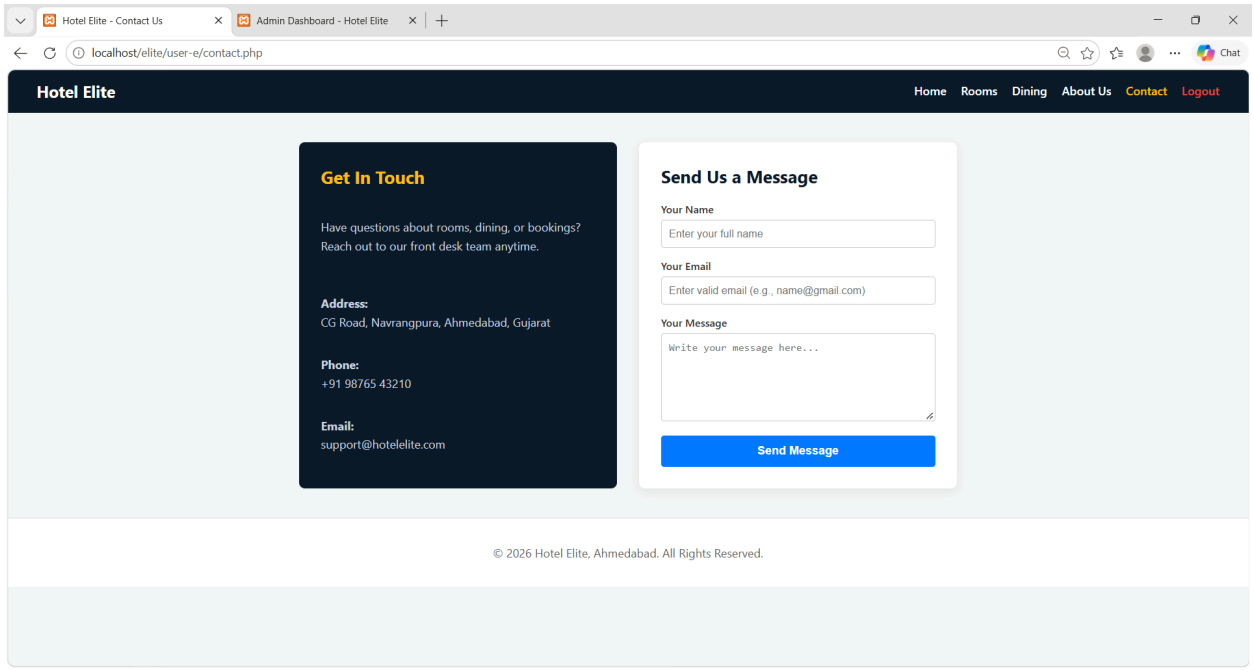



Hotel Elite - Dining
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Veggie Burger
₹180
Add to Cart
White Sauce Pasta
₹280
Add to Cart
Healthy Green Salad
₹150
Add to Cart
Chocolate Brownie
₹200
Add to Cart
Your Cart
No items selected yet.
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Your Name
Enter name
Room Number
Enter room no
Payment Method
Cash / Pay to Room Bill
Order Food
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ABOUT US

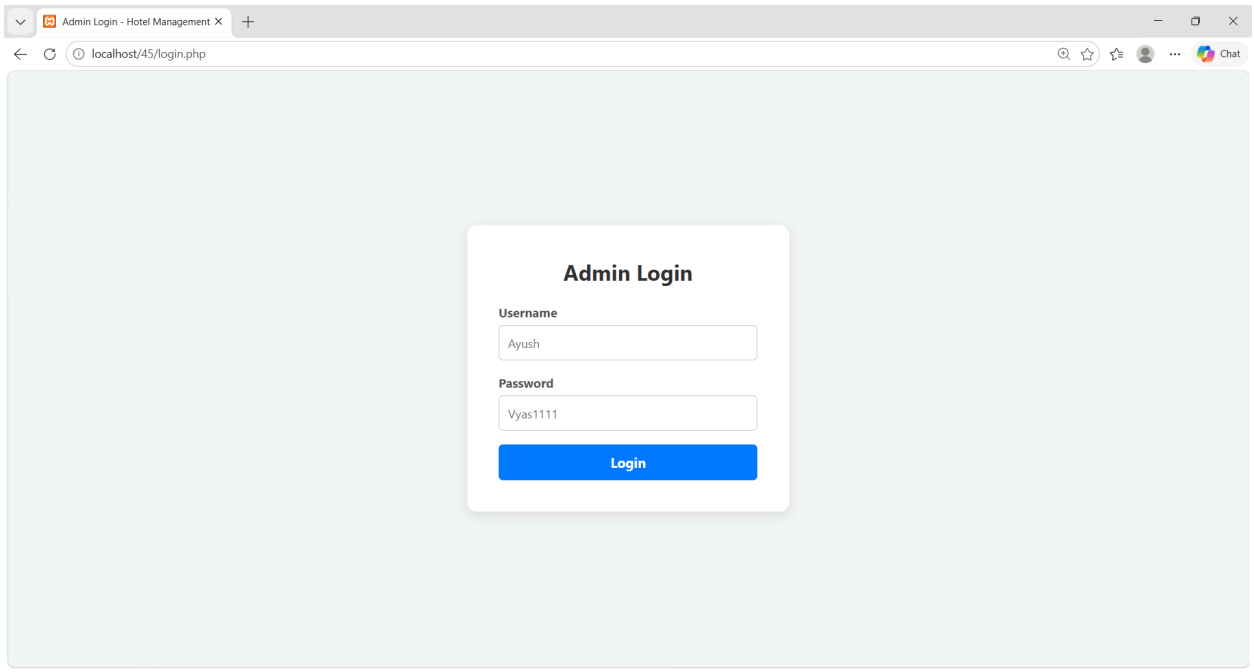

Hotel Elite - About Us
localhost/mahavir/about.php
Chat
Hotel Elite
Home
Rooms
Dining
About Us
Contact
Logout
About Hotel Elite, Ahmedabad
Welcome to Hotel Elite located in the heart of Ahmedabad, Gujarat. We combine modern luxury with traditional Gujarati hospitality to offer an unforgettable stay.
Our Location
CG Road, Navrangpura, Ahmedabad, Gujarat 380009
Chimanlal Girdharlal Road
Open in Maps
Old High Court
Navrangpura Rd
Bank Of India Society Rd
Municipal Market
Jadia Jewels Rd
Yanki Sizzlerr
Naturals Ice Cream
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CONTACT PAGE

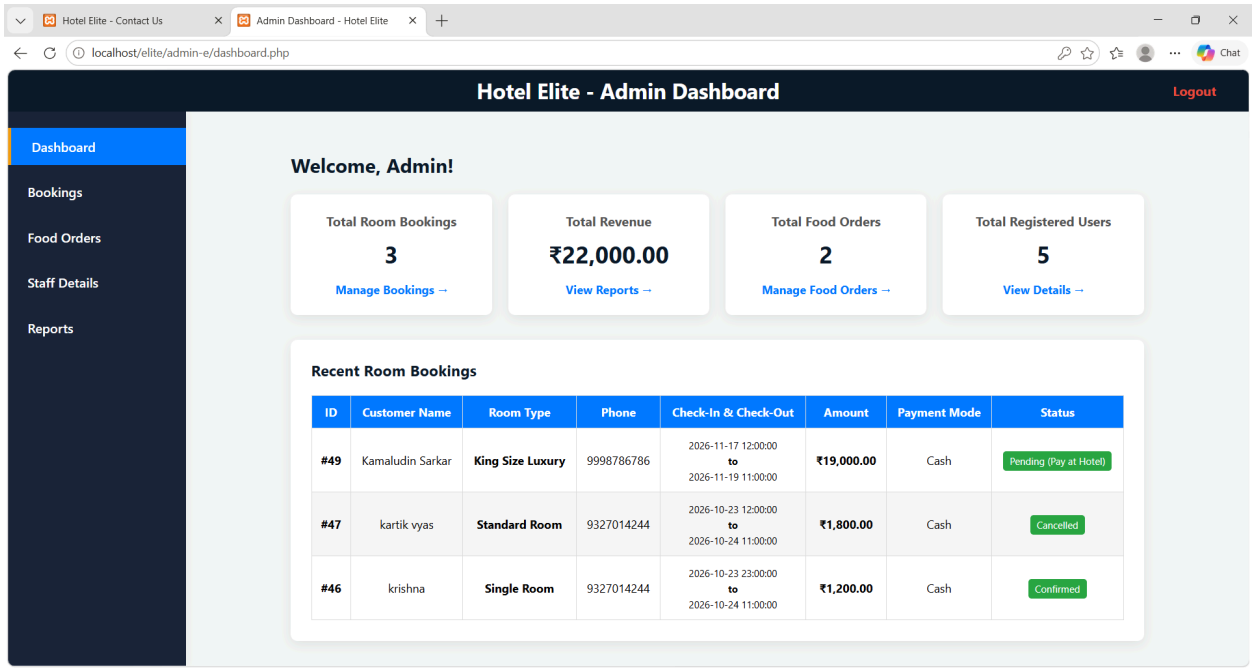


ADMIN PANEL

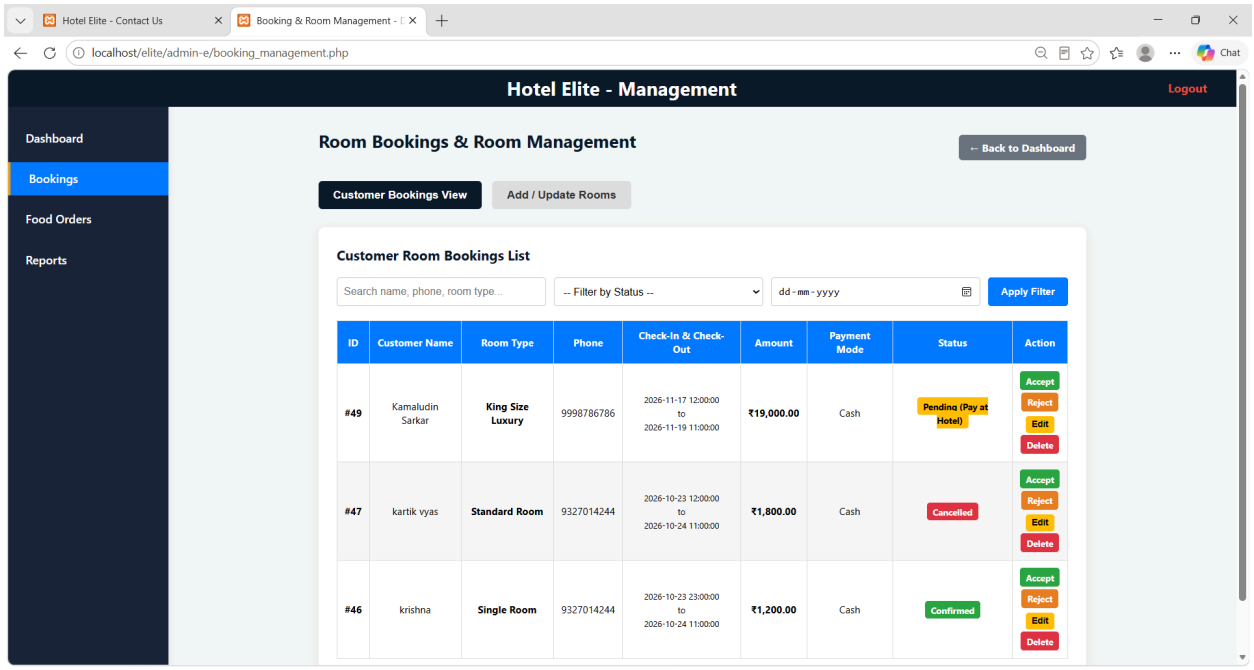
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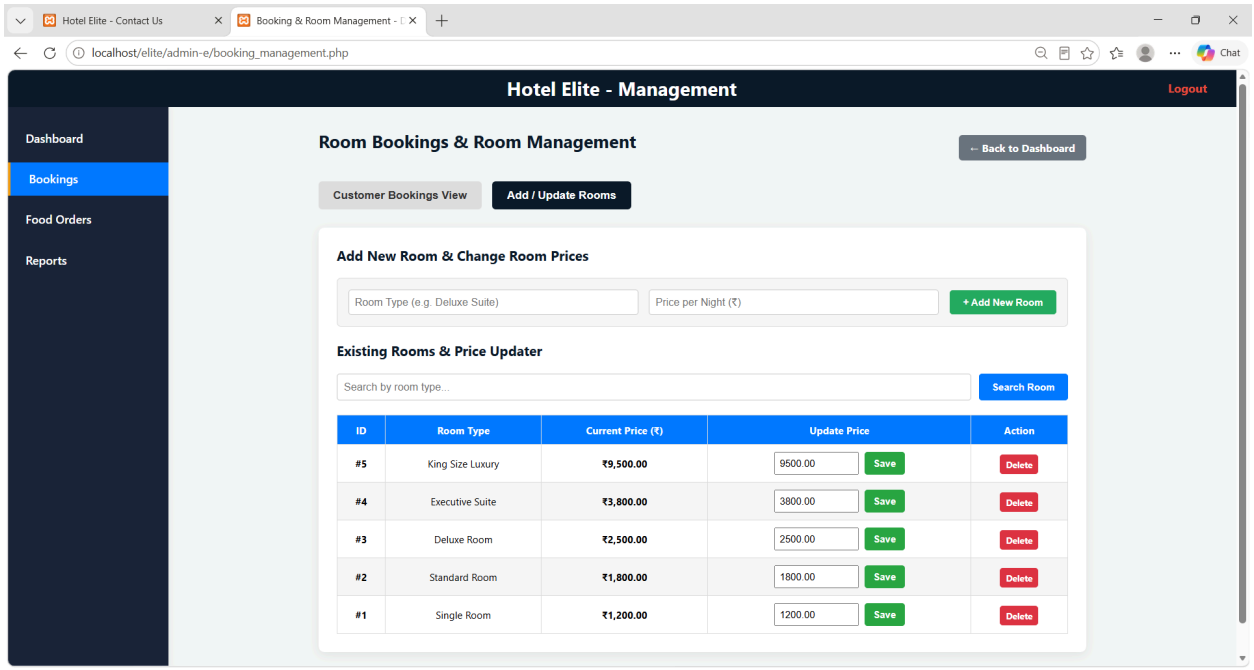
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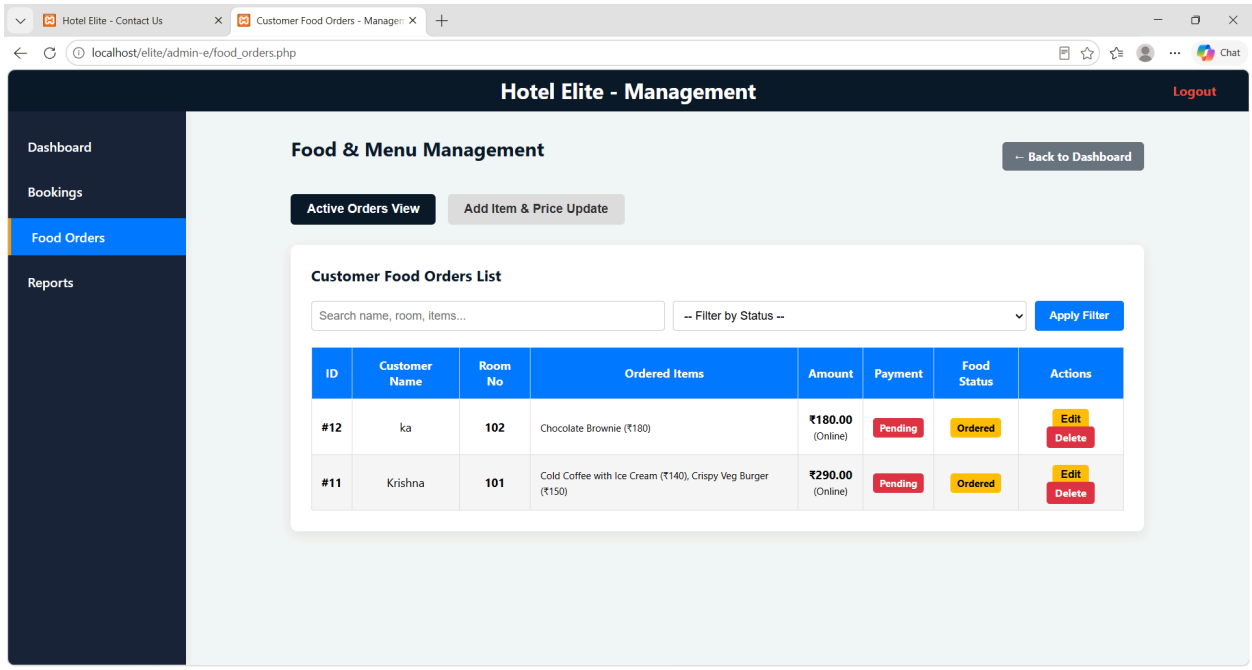
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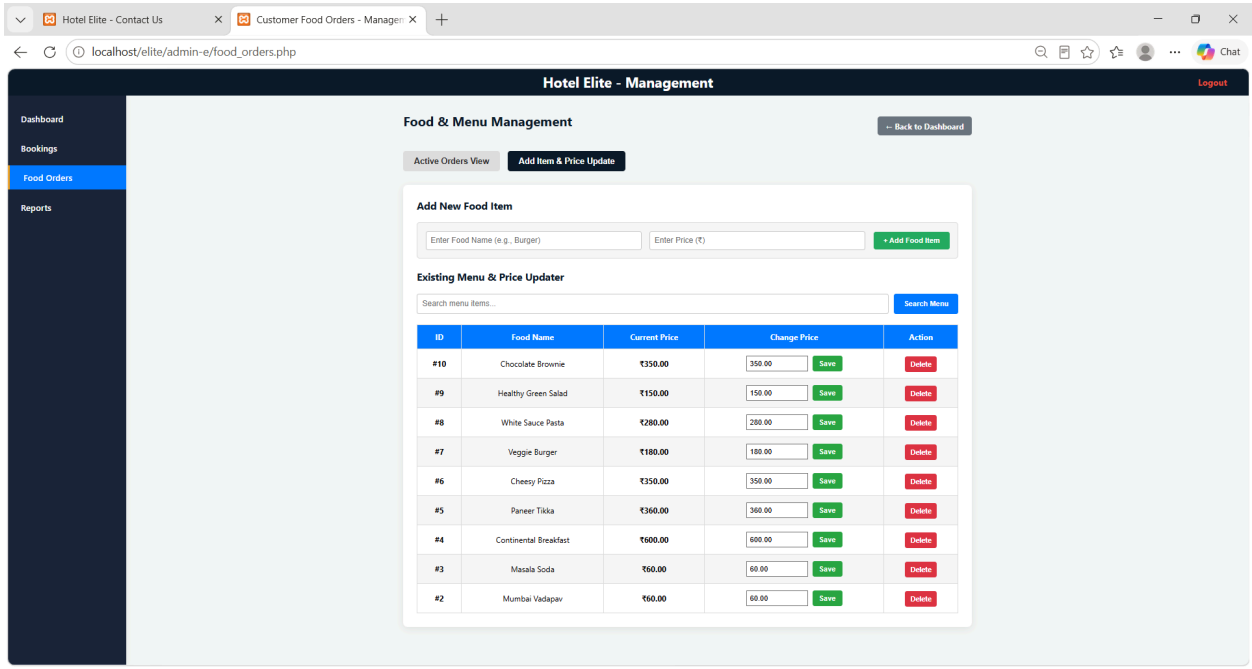
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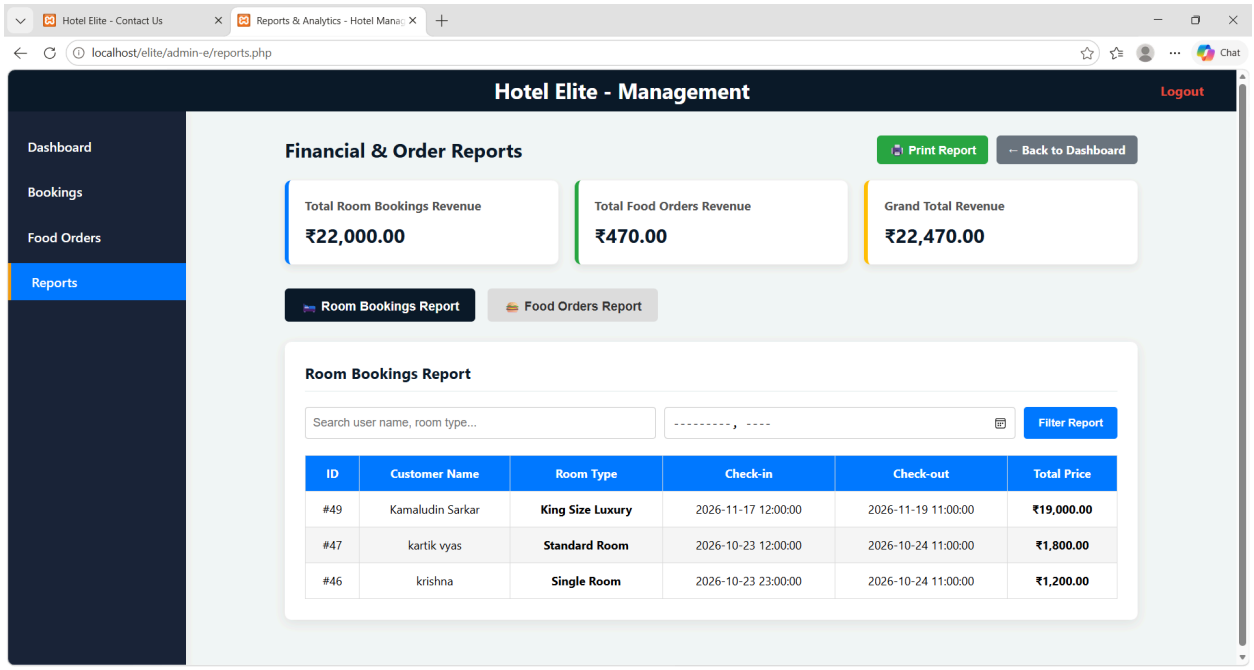
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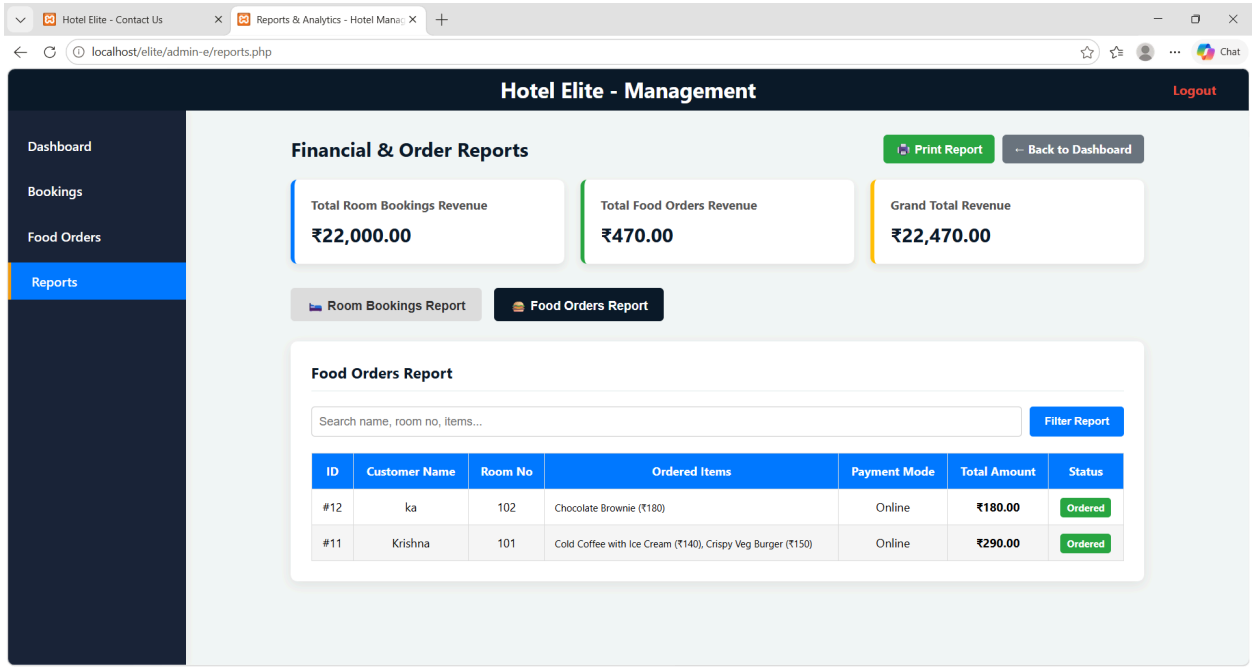
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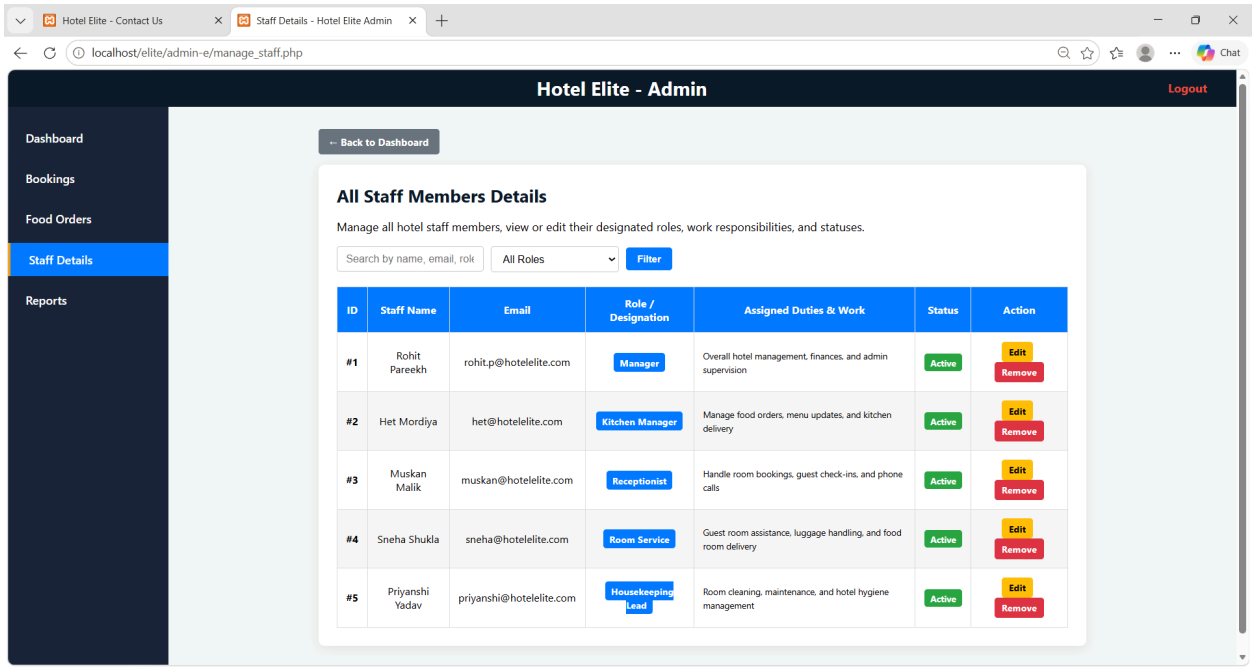
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Food_order_report



Staff_details



Users_details

